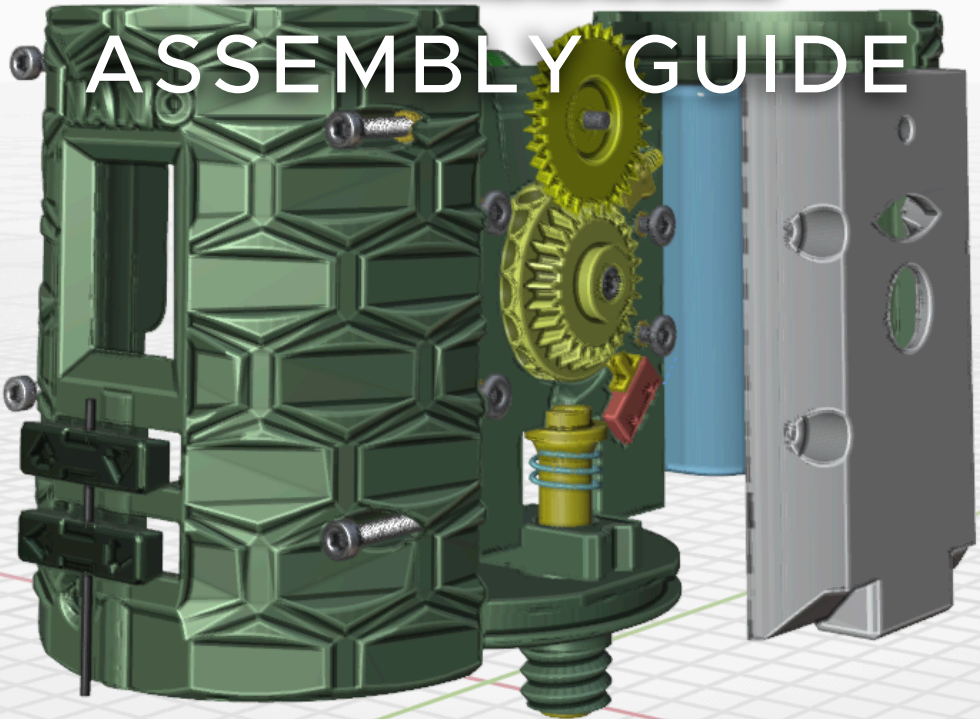


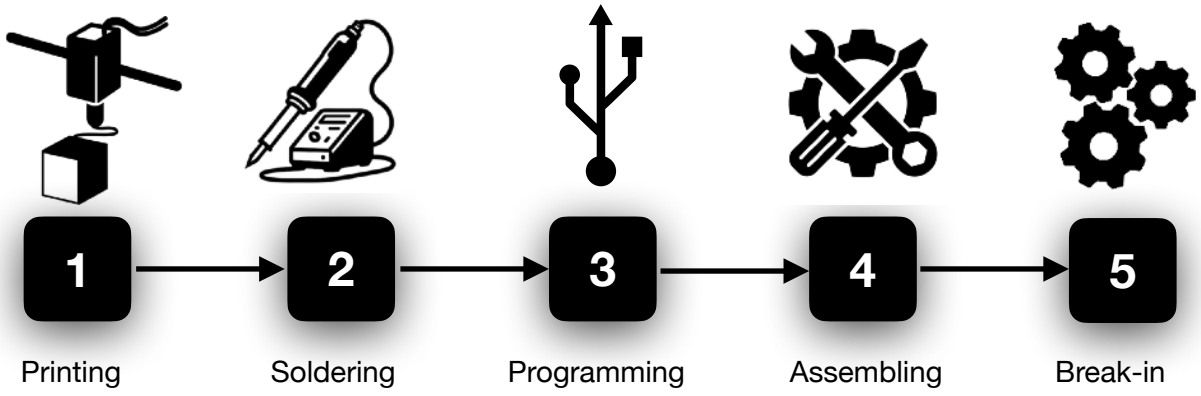
NANO

SpeedLoader

ASSEMBLY GUIDE



Summary



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Introduction



Thanks for downloading this product, This Assembly guide will show you the steps to assemble the NANO SpeedLoader,

In many ways this object will become your all time companion on the Airsoft field, while on the safe zone but also on the field due to his pocket side, you can carry it with you all day and reload your mags in few seconds.

The key advantages of this product is the integration of an ESP32 Ship, providing more customization options, language support, animations, and a motor controller board that allows users to adjust the speed for enhanced control and precision.

This project is developed with an evolving Arduino code, continuously enhanced based on user feedback. As an enthusiast this solution may have limitations from various perspectives. All enhancement feedbacks are welcome to refine and optimize the design.

This assembly journey will please all electronic and DIY enthusiasts, it takes fews days for the 3D prints, and few hours for soldering, so you should be able to put everything together few hours following this guide.

Packs and Options

The **Nano Speedloader** is offered in several versions to suit your needs, skills, and budget.
The **base pack** includes only the **3D design files**, the **components list**, and this **assembly guide**. You will need to purchase and assemble all parts yourself.

ℹ️ **Note:** In all options, the 3D-printed parts must be produced DIY using provided design files (STL archive)

Nano Speedloader – Pack Comparison Table

Find all our products on **https://protoforge.fr/**

Feature / Option	Cults3D Base Pack	Naked PCB	Tactical Pack (1)	Complete Pack (2)	Mission Pack (3)
3D Design Files & Assembly Guide	✓	✗	✗	✗	✗
Official PCB board	✗	✓	✓	✓	✓
Hardware and Electronic. Components Included	✗	✗	✓	✓	✓
ESP32 Included & Pre-Programmed	✗	✗	✗	✓	✓
Servo Motor Included	✗	✗	✗	✓	✓
Battery Included	✗	✗	✗	✓	✓
Components Pre-Soldered	✗	✗	✗	✗	✓
Buyer Needs to Solder Components	✓	✓	✓	✓	✗
Buyer Needs to 3D Print Housing	✓	✓	✓	✓	✓
Suitable For	DIY experts who source all parts themselves	DIY experts who want minimum soldering	Makers wanting a budget-friendly kit	Buyers wanting all electronics ready for soldering	Buyers wanting fast assembly with minimal work

Project Time :

The Nano Project is estimated to take **32 hours,** some accelerators like the **advanced packages** will reduce significantly the **time and complexity** on soldering, Programming and assembling phases.

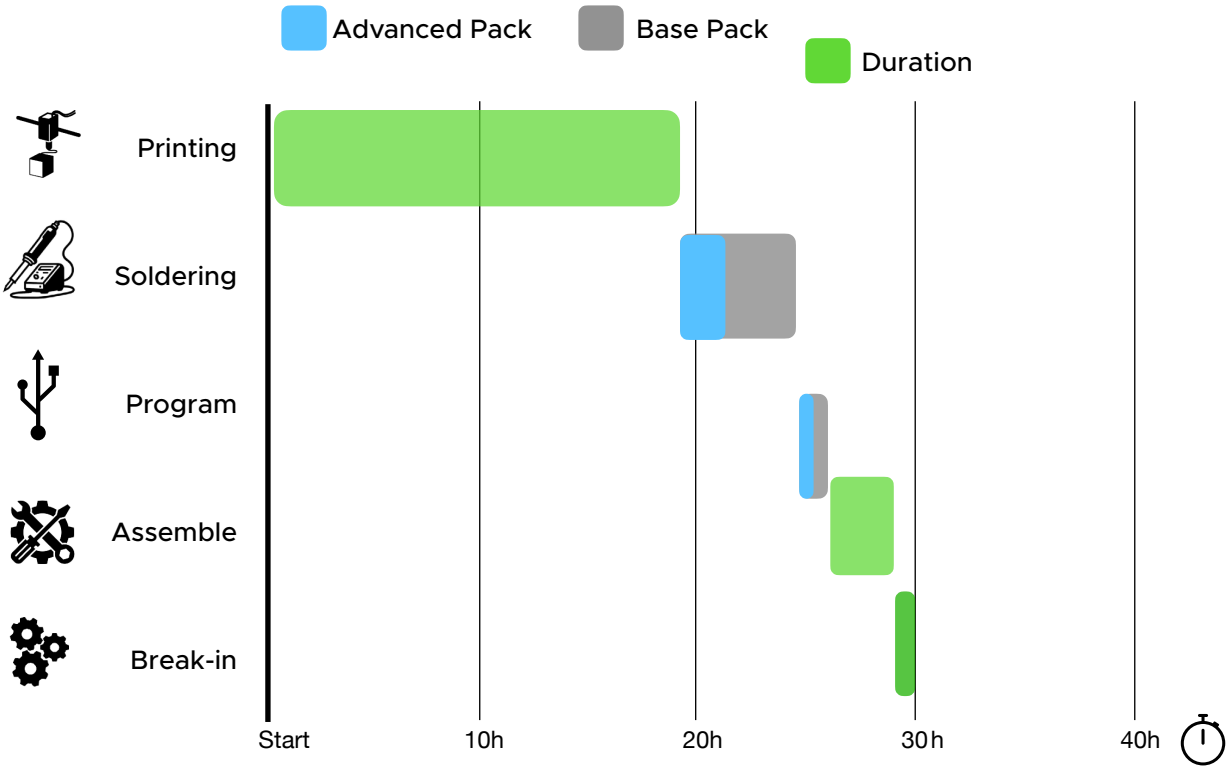
Printing : time estimation of **21 hours** is based on a **Creality K1** standard print time using **recommended parameters** and materials.

Soldering : it take less than **1 hour** to finish soldering using **advanced package .** Full DIY approche might take **from 5 hours to 12 hours** depending on your soldering skills.

Programming: Programming might take between **30min to few hours**. Some of the packages comes with a **preprogrammed TTGO** Board.

Assembling: Depending on the time spent on the preparation, this last building phase will take from **1 to 2 hours**.

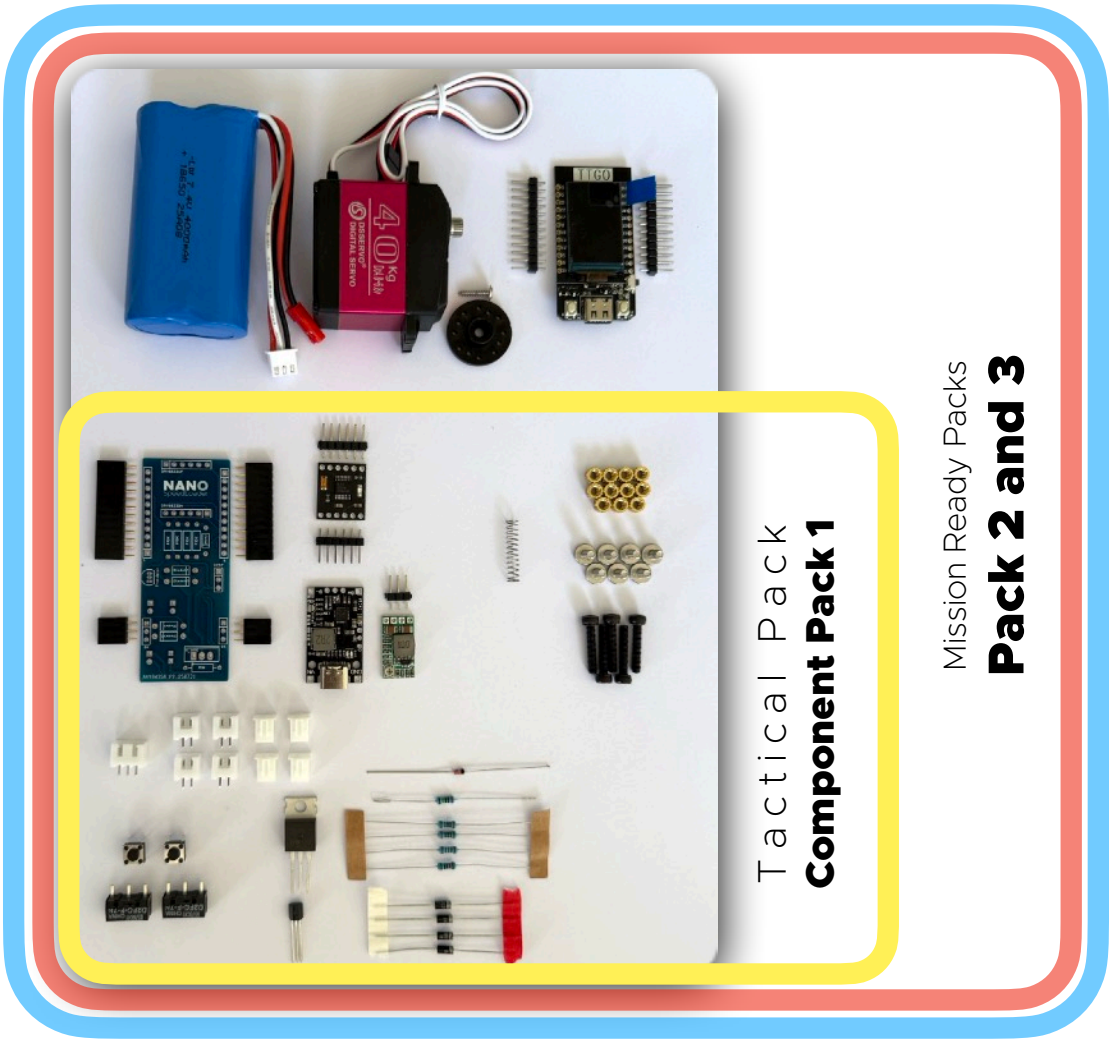
Break-in : This phase can be stretched over time, the idea is to run the gears and make sure everything is running smoothly. from **1 hour.**



Requirements & Prerequisites

Necessary Tools:

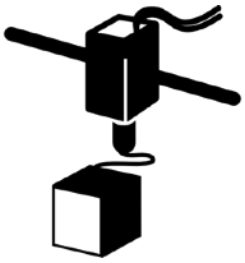
- Standard screwdriver set
- Hex keys
- Soldering iron
- Hot Glue Gun / Superglue
- Multimeter
- Tweezers
- Pliers
- Hobby Knife
- 2mm Drill bit
- Sand Paper



Component list

Pack N°	Component	Reference	Quantity
1,2,3	USB 2S LiPo BMS 15W	www.aliexpress.com/item/1005010507492400.html	1
1,2,3	Spring 0.4x4x10mm	Recovered from empty ball pen	1
1,2,3	Polycarbonate Sheet	Custom polycarbonate shield 25x35x2 mm	1
1,2,3	Spring 0.7x10X15mm	https://a.aliexpress.com/_EzfsTHm	1
1,2,3	Prototyping PCB 3x7cm	https://a.aliexpress.com/_EGaWbz0	2
1,2,3	Mosfet P - (IRF4905)	www.aliexpress.com/item/1005004336080925.html	1
1,2,3	Male headers 3 Pins	https://a.aliexpress.com/_EluV39Y	3
1,2,3	Female Headers 12 Pins	https://a.aliexpress.com/_EITBap4	2
1,2,3	Female Headers 3 Pins	https://a.aliexpress.com/_EITBap4	2
1,2,3	Connector JST 2/3 Pins	https://a.aliexpress.com/_Evrhc2O	4
1,2,3	M3 Threaded inserts	https://a.aliexpress.com/_EGBXgl4	11
1,2,3	Push Button - 5mm high	https://a.aliexpress.com/_EJfdT0O	2
1,2,3	M3 Hex Screw flat/large head (M3x6mm)	https://a.aliexpress.com/_EJFN6H8	7
1,2,3	M3 Hex Screw (M3x16mm)	https://a.aliexpress.com/_EvKJmyE	5
1,2,3	Microswitch (D2FC-E-7N)	www.aliexpress.com/item/1005011601223613.html	2
1,2,3	Transistors NPN (S8050 or 2N2222)	https://a.aliexpress.com/_EQUFVFw	1
1,2,3	Zener Diode (1N4739A)	https://a.aliexpress.com/_EIRX6IQ	1
1,2,3	Diode (RL207)	https://a.aliexpress.com/_EvhpqbW	4
1,2,3	Resistance (10KΩ)	https://a.aliexpress.com/_EvFshW6	4
1,2,3	Resistance (1MΩ)	https://a.aliexpress.com/_EvFshW6	1
1,2,3	Motor Driver Module (DRV8833 Black)	www.aliexpress.com/item/1005007987485248.html	1
1,2,3	Step Down Power Supply Module	https://a.aliexpress.com/_EJTJ2Qi	1
1,2,3	Pre-clamped XH-JST2 wires/connectors	www.aliexpress.com/item/1005010216985735.html	4
2,3	Recommanded Servo	www.aliexpress.com/item/4001006734924.html	1
2,3	Battery 7.4V 2S	www.aliexpress.com/item/1005008672212411.html	1
2,3	Esp32_TTGO_T-Display	www.aliexpress.com/item/33048962331.html	1

1 3D Printing 21 Hours

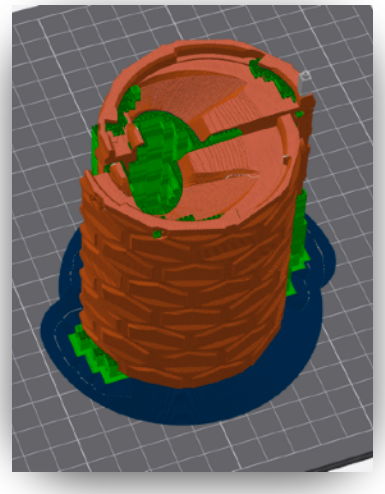


In this section, you will find all the necessary information to successfully 3D print the parts required for your Nano Speedloader. We will go through the recommended print settings step-by-step to ensure optimal strength, precision, and finish. The parameters covered include **layer height** (print quality), **number of walls** for durability, **support usage** for overhanging features, **recommended filament material** for best performance, and **ironing** to achieve a smoother top surface. Following these guidelines will help you produce parts that fit perfectly and stand up to regular use.

ℹ️ Note that the STL files are provided in the recommended print orientation.

- **Main Housing Front and Back** :





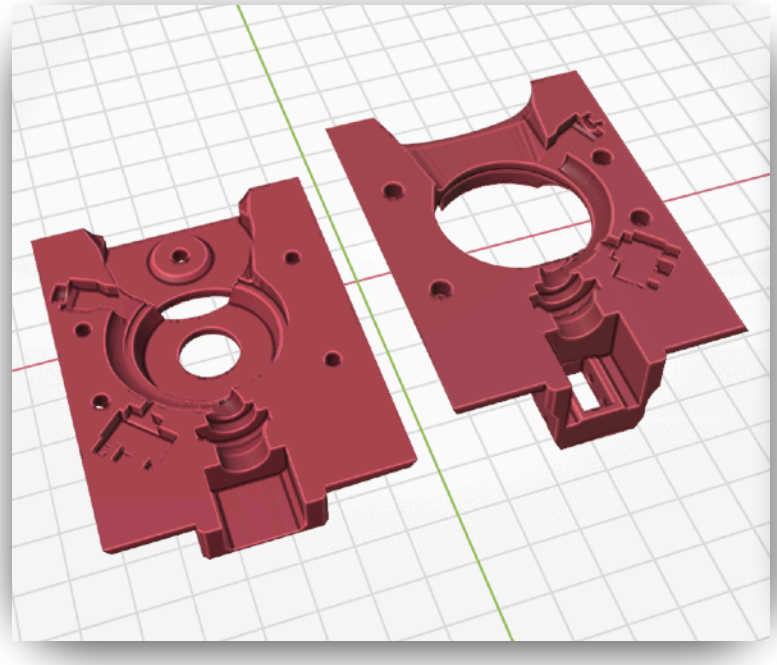
The main housing is the most prominent and visible part of the Nano Speedloader. For the best visual appearance and fit, it is recommended to print this component at high quality (low layer height). Due to its complex geometry, support structures are required to ensure proper printing of overhangs and intricate features.

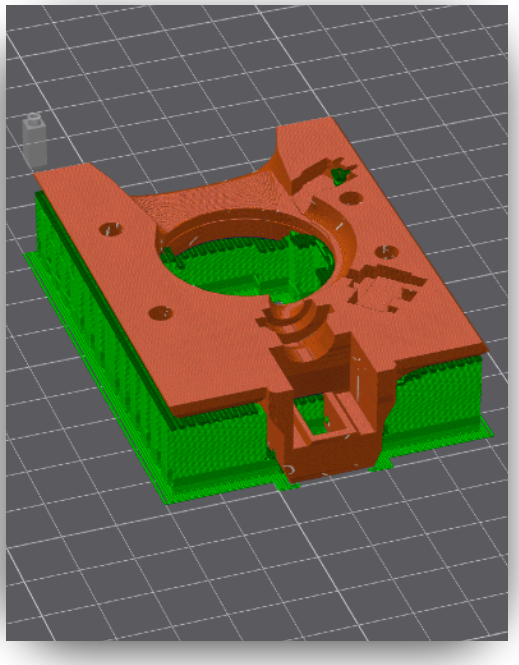
The estimated print time for the main housing is approximately **7 hours** when using the recommended settings. To optimize efficiency, it is advisable to print both housing halves in a single job, which will reduce total print time compared to printing them separately.

Printing Parameters :

Quality	Supports	Walls	Ironning	Filament
0.08 to 0.1 mm	Mandatory	2	Optional	ABS / PETG

Gearbox Parts A & B:





The gearbox shells are printed in **two parts** with a complex geometry, forming the structural backbone of the Nano Speedloader. Because this component must withstand significant mechanical stress during operation, it is strongly recommended to use an **infill density of 50%** to ensure sufficient strength and durability.

Print orientation is critical: the **inner side of the gearbox should face upward** to achieve the best possible surface finish and to maintain precise internal tolerances. Additionally, if your printer supports it, enabling the **ironing feature** will further improve the top-layer quality, ensuring smooth surfaces where the internal mechanisms operate.

Printing Parameters :

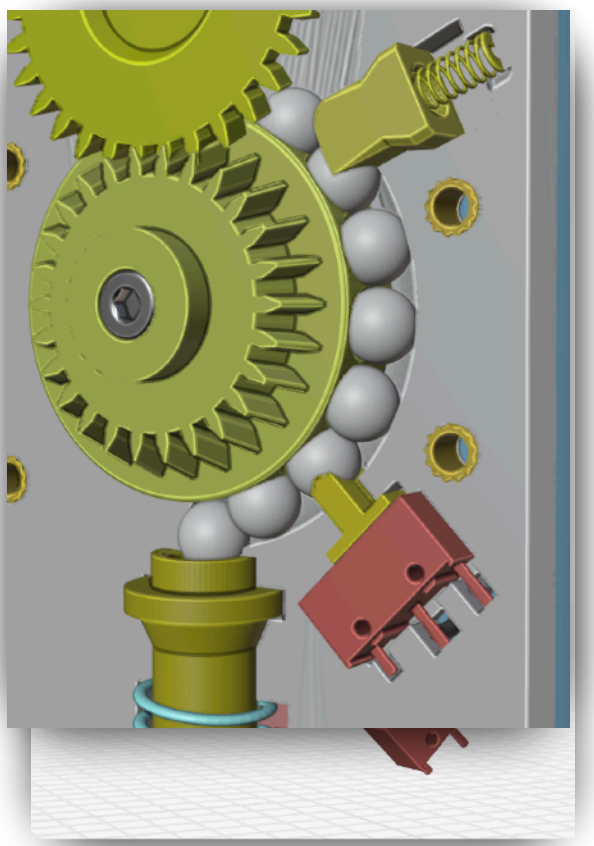
Quality	Supports	Walls	Ironning	Filament
0.2 mm	Mandatory	4	Required	Any

The components:

Few mobile components are integrated to the gearbox in order to fix some issues present in the original model

Stir wheel : The stir wheel is designed to improve the overall reliability of the Nano Speedloader by preventing BBs from getting stuck above the feeding wheel. Without this component, it is common for BBs to cluster together, which can block the feeding mechanism and require the user to shake the speedloader to clear the jam.

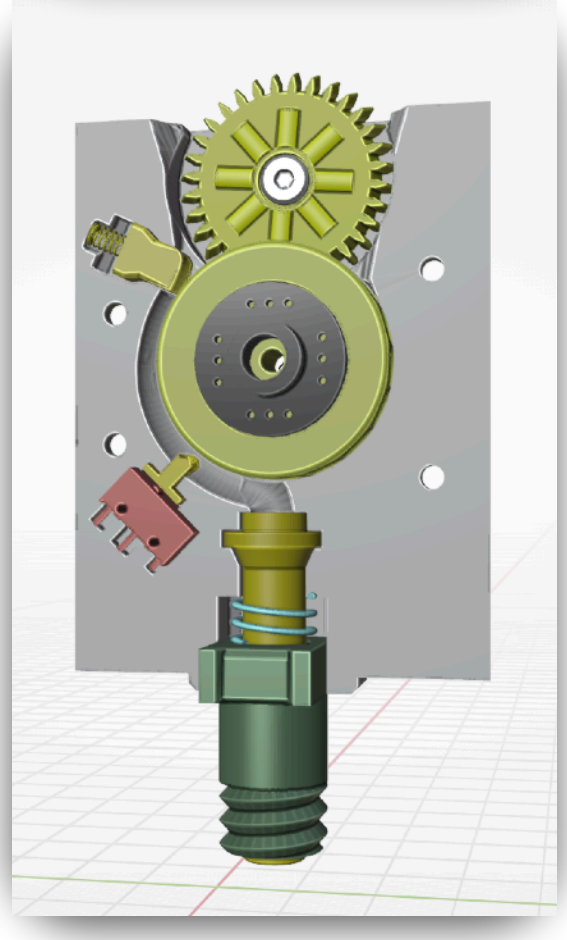
By continuously stirring the BBs during operation, this wheel helps maintain a steady flow of ammunition and minimizes the risk of feeding interruptions. This simple yet effective addition ensures smoother operation and reduces the need for manual intervention.



Anti-Pinching Knob : The anti-jam knob is designed to prevent BBs from being improperly engaged on the feeding wheel. Occasionally, a BB may get pinched between the wheel teeth and the gearbox wall. This can cause BBs to be ejected uncontrollably and, more critically, accelerate wear on the thin edges of the feeding wheel.

To solve this, the anti-jam knob is spring-loaded to create a flexible interface that absorbs misalignments and allows jammed BBs to be released safely. The spring used here is a **repurposed pen spring**, with a free length of **12 mm** and a fully compressed length of **4 mm**. The spring attaches directly to the knob via a tight-fit hole, while the opposite end locks onto a **dedicated pin** located inside the gearbox housing.

This simple mechanism greatly extends the lifespan of the feeding wheel and improves the overall reliability of the Speedloader.



Internal Counter Micro-Switch : The internal counting switch knob serves as the interface between the BBs and the switch mechanism. Its role is to absorb friction and wear, while remaining easy to replace if needed. The part has **no specific orientation** and can be installed in either direction during assembly.

Since this product is still new, long-term durability feedback is limited. However, testing on the first prototype has shown excellent performance, with over **200,000 rounds** fired while the knob remains fully functional. That prototype was printed in **ABS**, which provided good resistance to wear. Depending on the filament you use, durability may vary. We recommend **periodically inspecting the knob**, particularly if you notice any counting inaccuracies, as this may be an early sign of wear.

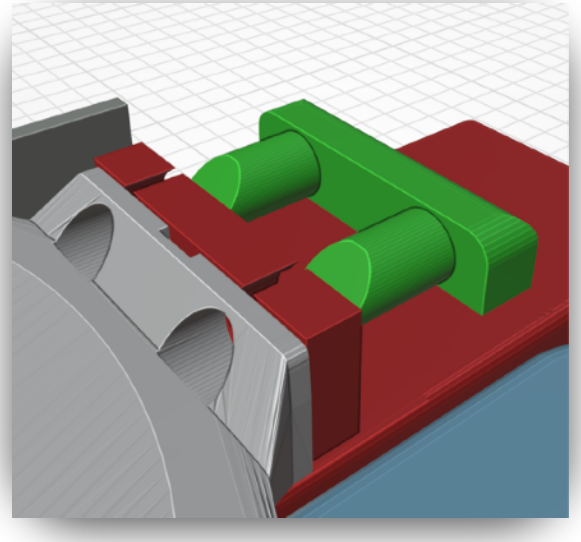
Feeding wheel : The main feeding wheel is a critical component of the Nano Speedloader, as it directly interacts with the BBs and transfers rotational movement

from the servo motor. The wheel incorporates an integrated gear that allows the servo's torque to be efficiently transmitted to the stir wheel.

The design of the feeding wheel has been optimized so that BBs make only minimal contact with its surface. This ensures that each BB pushes the next one into the feeding channel, rather than relying solely on the wheel's geometry to maintain and move the BBs. This improvement significantly reduces the mechanical stress on the wheel, particularly when the magazine becomes nearly full and requires greater force to continue feeding.

You will find in the Archive 3 variants depending on the servo horn you are using and the type of servo.

Servo Clip : The servo motor normally requires four screws for secure mounting. However, since the Nano Speedloader's housing already provides strong support, only the two lower screws are necessary. To simplify assembly and ensure proper alignment, a servo clip has been added.



Once the bottom screws are in place, the clip is inserted into the upper mounting holes of the servo. The part is designed as a tight fit, ensuring stability without loosening over time. This solution reduces assembly effort while maintaining a secure and durable attachment of the servo motor.

Printing Parameters :

Quality	Supports	Walls	Ironning	Filament
0.1 mm	Optional	>4	Optional	ABS/PETG

Nano Extension (Reservoir) and Top Cover: The Nano Extension is an optional reservoir that increases the BB capacity of the Speedloader to approximately **1,300** BBs. For users who prefer a more compact setup, the extension can simply be omitted.



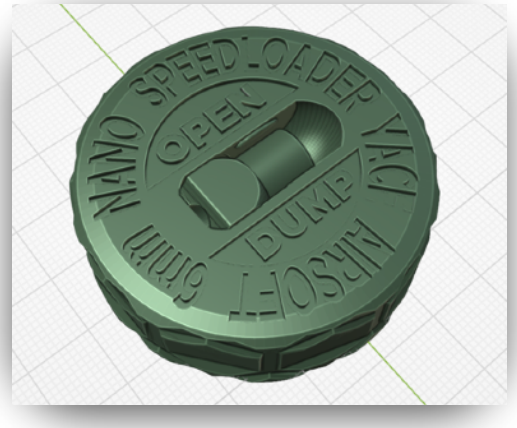
The Top Cover is designed to fit in both configurations:

With the extension: it seals the reservoir to maintain maximum capacity.

Without the extension: it attaches directly to the Nano Speedloader, resulting in a lower-profile design with minimal built-in BB capacity.

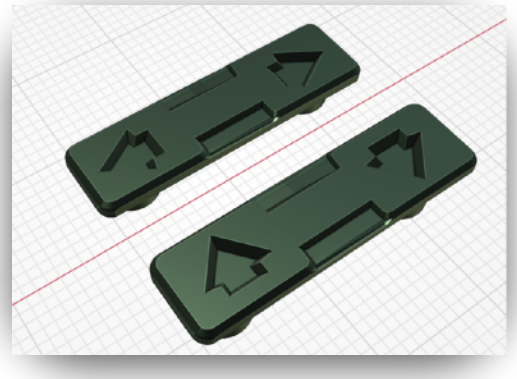
In this second configuration, the cover's screwed neck allows the attachment of a standard 3,300 BB ASG bottle, which can be used to feed BBs directly into the gears. This offers flexibility between a compact, portable setup and an extended high-capacity configuration, depending on the user's needs.

The Cap: is the final external piece that closes the Nano. It includes a built-in dump/unloading pipe that can be folded away when not in use.



A click-lock mechanism ensures the pipe remains firmly in either the open or closed position, preventing accidental movement while still allowing smooth operation with the right amount of force. This feature is especially useful at the end of a game, when you may wish to quickly unload the remaining BBs and start fresh, regardless of how many are left in the reservoir.

Buttons: This last piece completes the design of the housing, they are fixed using a central rode, you can use few centimeters of filament and slide it from the bottom to secure the button on the front part of the housing.

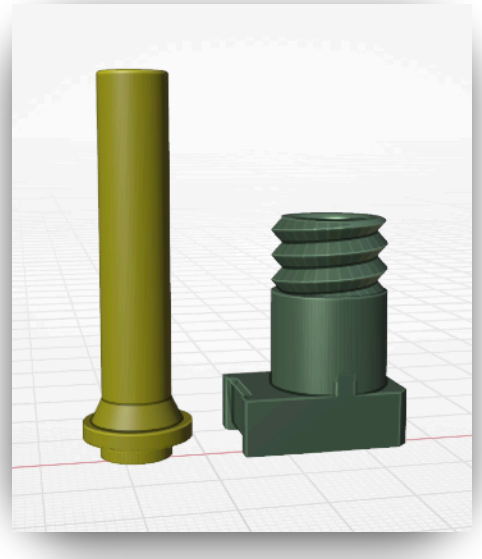


Up and Down buttons goes on top, **left and right** does on the bottom side.

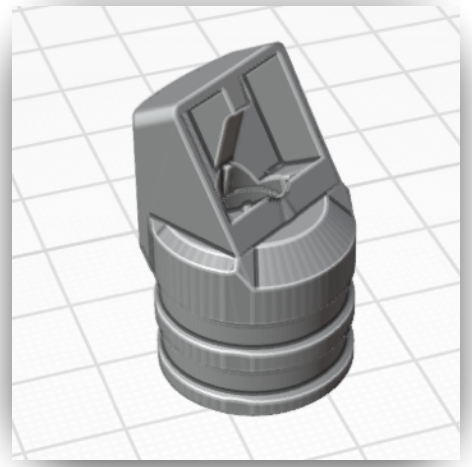
Printing Parameters :

Quality	Supports	Walls	Ironning	Filament
0.1 mm	Required	2-3	Recommended	ABS/PETG

Feeding Barrel : The feeding barrel is installed inside the gearbox before final assembly. Its design includes a slightly narrowed exit at the tip to ensure that remaining BBs stay inside once feeding is complete. This prevents accidental BB loss and keeps the loading process clean and reliable.



Trigger and Adapters : As its name suggests, the trigger is responsible for actuating the microswitch to start the feeding process. On the external side, the trigger features a threaded end, allowing the attachment of various feeding adapters. These adapters make it possible to accommodate special magazine types, such as Gas, CO_2 pistol, and GBB magazines.
The trigger is supplied together with these adapters, ensuring compatibility with a wide range of platforms.

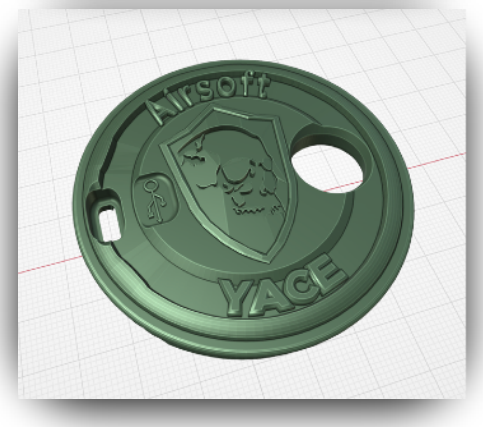


Lower Cover : The lower cover is the final piece completing the Nano Speedloader assembly. It has two primary functions:

- Securing the battery and BMS in place.

- Holding the trigger mechanism once its spring has been positioned.

This cover locks everything neatly within the housing, providing structural rigidity and ensuring that all internal components remain correctly aligned during operation.



Printing Parameters :

Quality	Supports	Walls	Ironning	Filament
0.2 mm	Recommended	2-3	Recommended	ABS/PETG

2 Soldering 1to8+ Hours



Watch these Youtube videos If you're new to soldering stuff:

https://www.youtube.com/watch?v=3jAw41LRBxU
https://www.youtube.com/watch?v=8kyo1PdkFLE

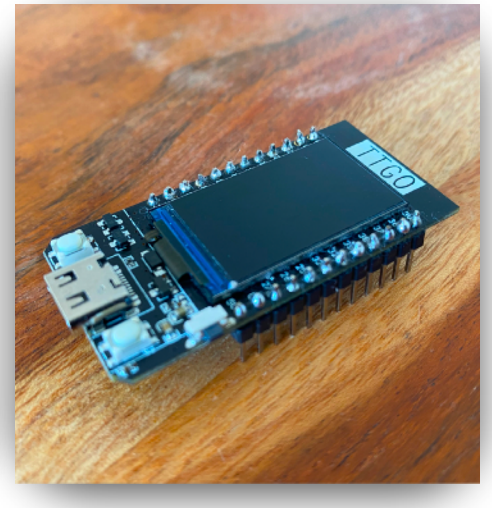
First step where you will have soldering to do is on the TTGO Board

The pins to solder on the board are included within the TTGO package

ℹ️ It is recommended to **apply some flux** to facilitate the soldering process, as the pins are really close, it will easily become a mess and can lead to shortage, thus the flux makes the process way easier.







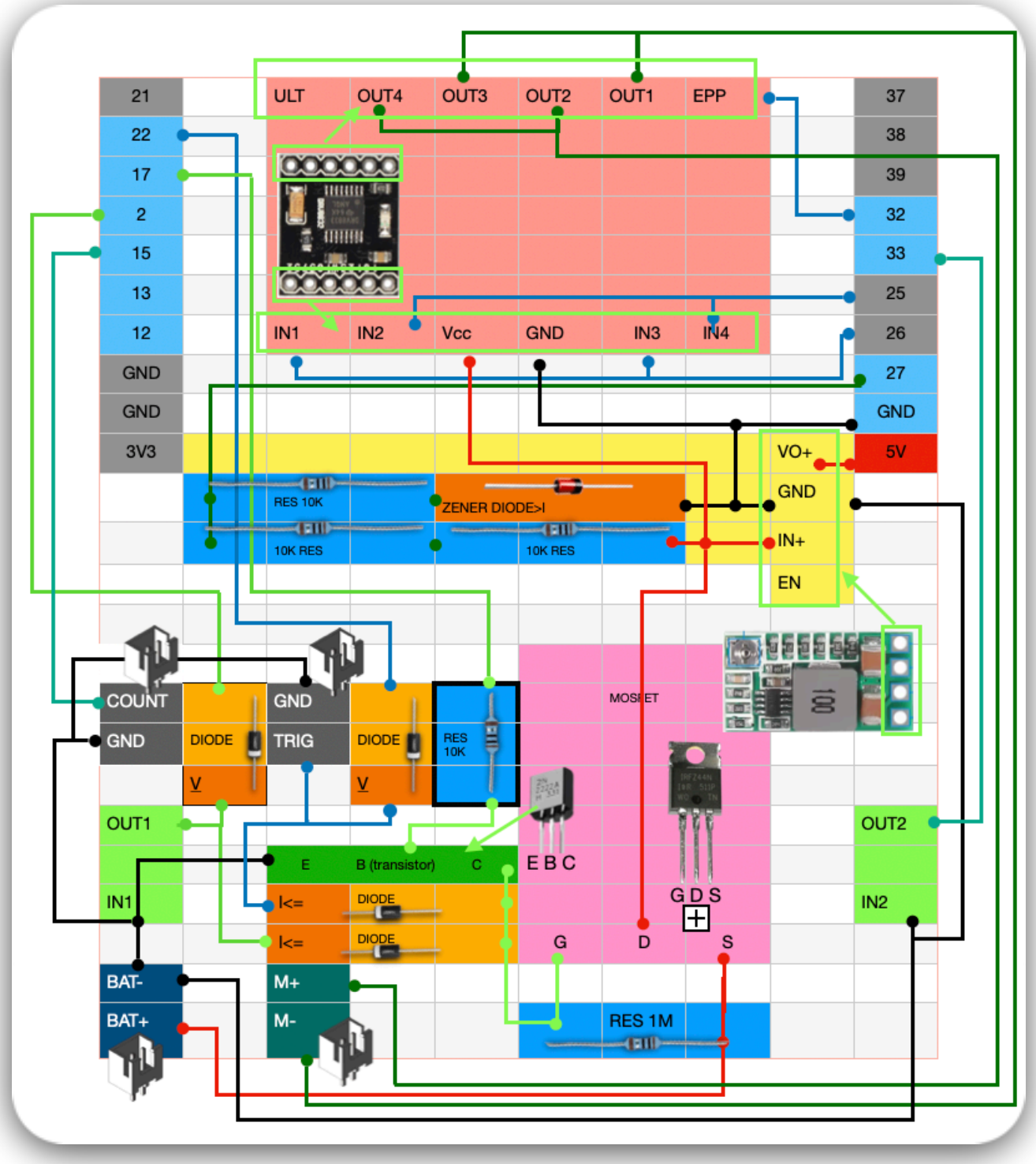
Once the soldering complete it's recommended to **clean the board** with a toothbrush and some **isopropyl alcohol**, the same used for cleaning your 3D printer plate bed to remove the glue.

The TTGO Board has 12 pins each side (24 total), only 11 pins are used (22 total). 2 upper pins are un-used (G & 3v)

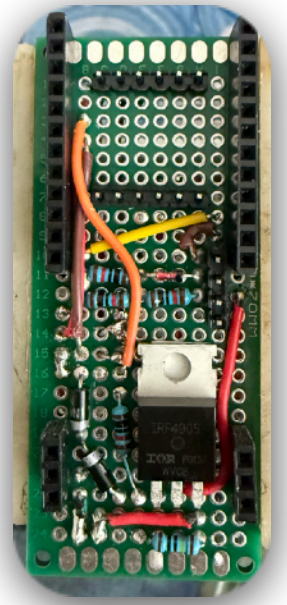
Wiring Diagram and PCB Assembly (Base Pack Only)

This section provides a detailed overview of the Nano Speedloader's electronic connections and the step-by-step process of assembling the PCB. To begin, a wiring diagram is presented, showing the relationships between all key components and the TTGO microcontroller. This diagram will serve as your main reference throughout the building process.

The PCB assembly is divided into three main layers, making the process easier to follow and reducing the risk of mistakes:



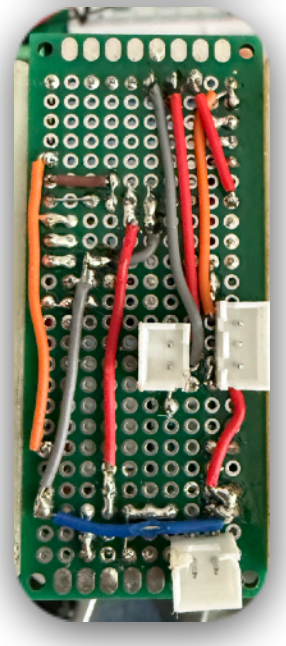
Layer 1 – Base Components



Solder the through-hole components (THC) and pin headers , these will provide the foundation for attaching other modules later on.

When soldering the through-hole components and headers, the **wires should primarily be soldered on the backside of the PCB**, as this provides a cleaner layout and prevents interference with the modules that will be mounted later. However, in some cases, it may be more practical to solder certain wires on the **front side** of the board. This approach can help reduce wire length or avoid congestion in tight areas. The accompanying picture highlights these exceptions, showing how front-side soldering can be used when necessary without affecting the final assembly.

Layer 2 – Wiring (Back side)

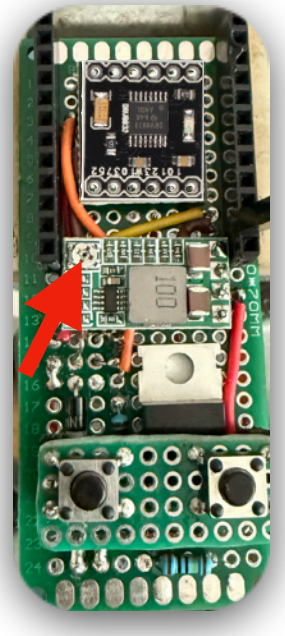


The first step on this layer is to solder the **JST connectors**, the setup requires **four (4x) JST-2 connectors**.

‼️ **Warning:** It is very important to place and solder these connectors on the **bottom-right side of the PCB (back side)**, as shown in the reference picture. The clearance inside the Nano Speedloader is extremely tight, so the connector **positions are predefined** and must be respected for proper assembly.

➡️ Once the connectors are firmly in place, proceed to solder the **wires** according to the wiring diagram. Take your time with this step, as neat and secure soldering will make the following assembly layers much easier and more reliable.

‼️ **Warning :** the **wiring diagram** is presented from front side, please refer to the wiring PDF document for **mirrored diagram**.



ℹ️ The reference picture shown may differ slightly from the wiring diagram, as it is based on an earlier circuit version. Please make sure to follow the wiring diagram, since it represents the latest and most accurate design

Layer 3 – Modules and Controller

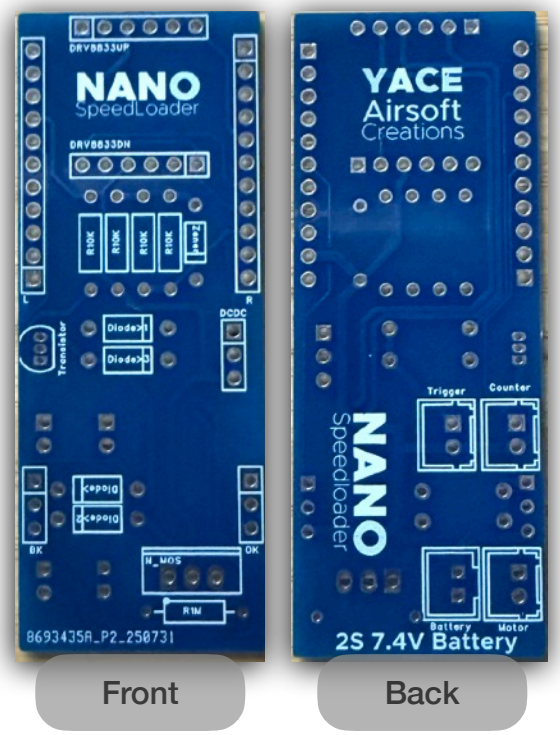
Before starting soldering please make sure to test proper functioning of all previous layer components as they will become inaccessible.

➡️ Install the external modules, such as the motor driver and DC-DC converter;

➡️ The DC-DC converter is soldered above the resistors, make sure that both layers are isolated with some isolating tape.

PCB Assembly (Advanced Packs)

All advanced packages includes a printed circuitboard to level up the product quality and reliability, the assembly consists of soldering the components in the following order



1) (Front) Components soldering

- **4x** 10k Resistor, **1x** 1M Resistor, **1x** Zener Diode, **4x** Diode, **1x** Mosfet, **1x** Transistor

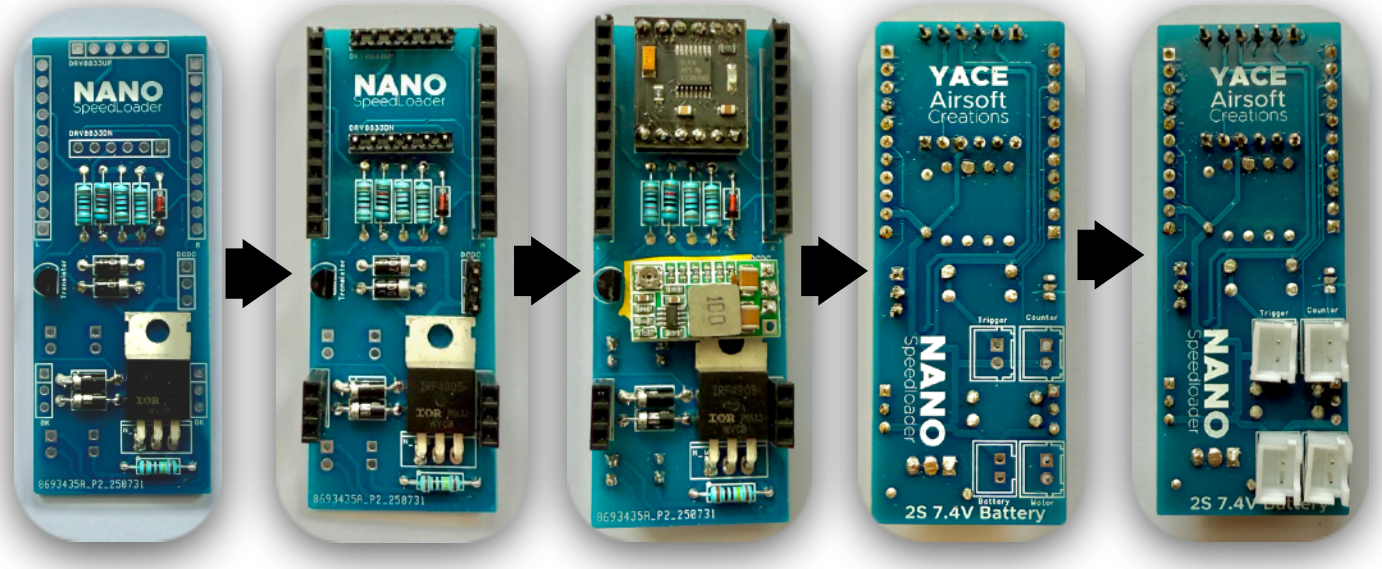
2) (Front) Pin Header soldering

- **2x** 11pin F-Header**, 2x** 3Pin F-Header**, 2x** 6pin M-Header**, 1x** 3pin M-Header

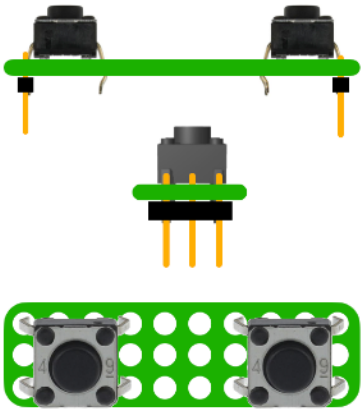
3) (Front) Modules soldering (DC-DC converter + motor controller) : apply some isolation tape on backside of the regulator to avoid shortcut

4) (Back side) Cleaning up the wire excess : Removing excess wire of THC components and pin headers to have the board surface clean.

5) (Back side) Connector soldering : Connectors has to be placed on the trace, polarity is critical.



Buttons Module :

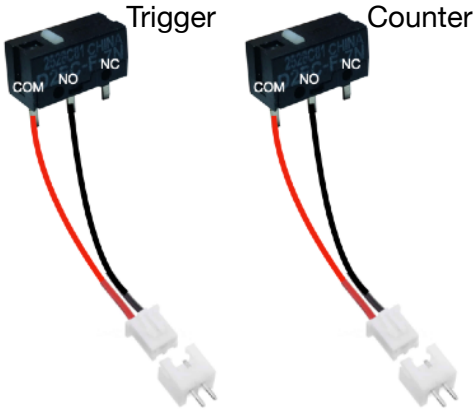


To assemble the control buttons, you will need a small piece of prototyping board measuring **3 × 10 holes** (approximately **26 × 8 mm**).

- Place the button pins at both extremities of the board.
- On the opposite side (back of the board), install a **3-pin male header** in the last row. If you are using a **standard prototyping board**, the button pins and the male headers can share the **same hole**, simplifying the layout.

ℹ️ Only **two of the pins** will be used for the connection, but keeping the middle pin in place adds rigidity and strength.

Microswitches :

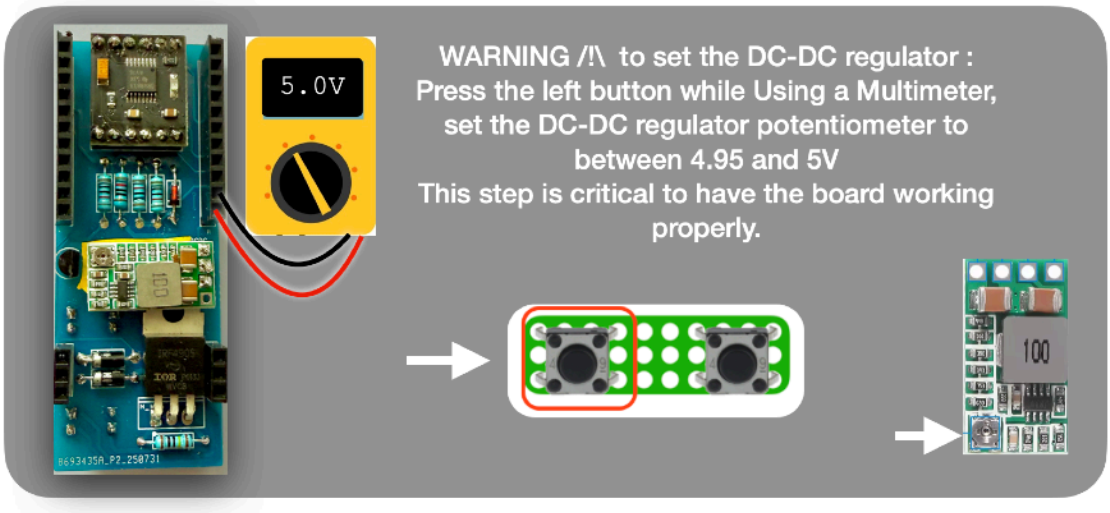


The microswitches are essential to the operation of the Nano Speedloader:

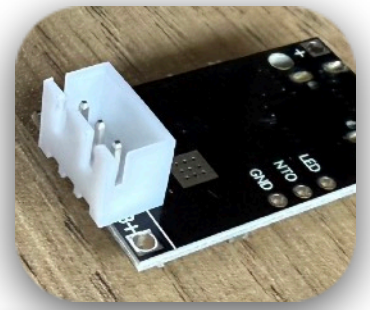
ℹ️ The counter switch wires must be soldered **before final assembly**.

➡️ After soldering, slide the wires through the **exit hole** in the gearbox shell (check assembly section for more details).

ℹ️ The wires recommended length: 10cm.

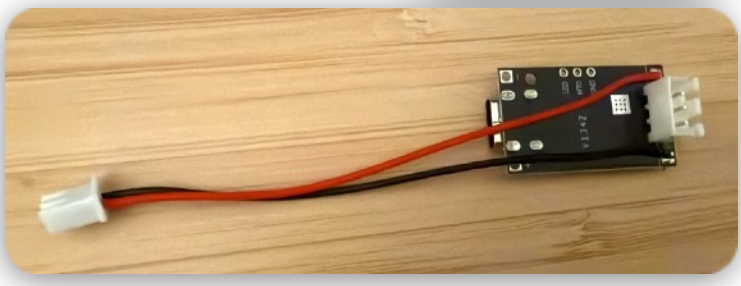


BMS Module :



The BMS module is in charge of managing the battery cells levels, this model by default set up to manage **7.4v 2s Li-Po Battery**, it requires a **3pin JST** connector to be soldered on the backside of the board, the orientation of the connector is critical as the IC will blow up if the polarity is switched, thus ensure you place the connector is the orientation shown in the picture.

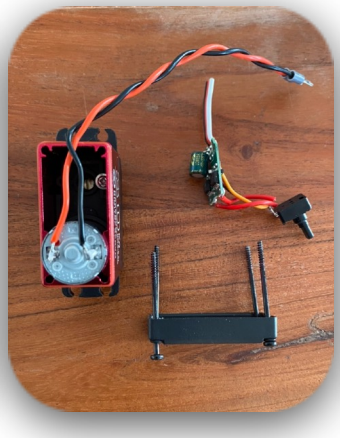
Finally, solder the power cable that will feed the circuit from this board. Make sure to follow the polarity sign to avoid fraying the board.

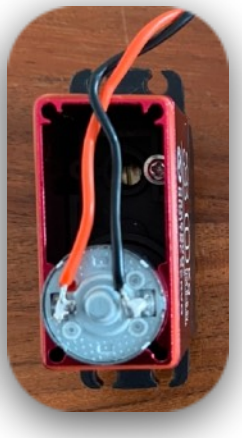


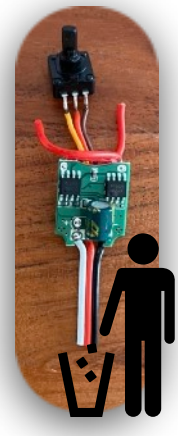
Servo-Motor preparation :

The servo wiring need to be simplified to remove the internal board and potentiometer as they will become useless, in this project the Servo. Is used as regular motor, you can solder a regular wires to (+) and (-) wires will be connected to the board.







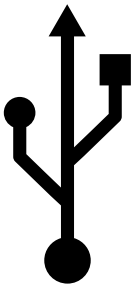


The Battery:

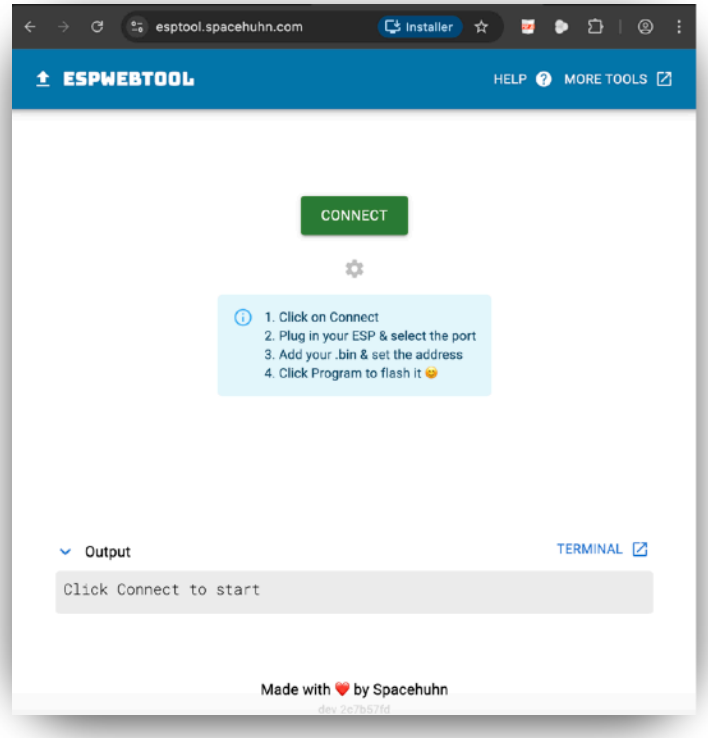
In order to have the Speedloader working properly, it is required to use LI-PO 7.4V 2S battery, as the space inside the Speedloader is very tight, you need to use the reference recommended or an equivalent.



3 Programming 10 Min



The Board used on this project is offering Over The Air (OTA) firmware update, meaning you can update the firmware and add the new features using WIFI, without the need of a computer, you will only need your phone for that.



However, the **first upload** need to be performed via **USB Cable**, made easy using the **ESP Web Tools**.

Using either **Chrome** or **Edge** Browsers,
➡️ Go to the following url :

https://esptool.spacehuhn.com/

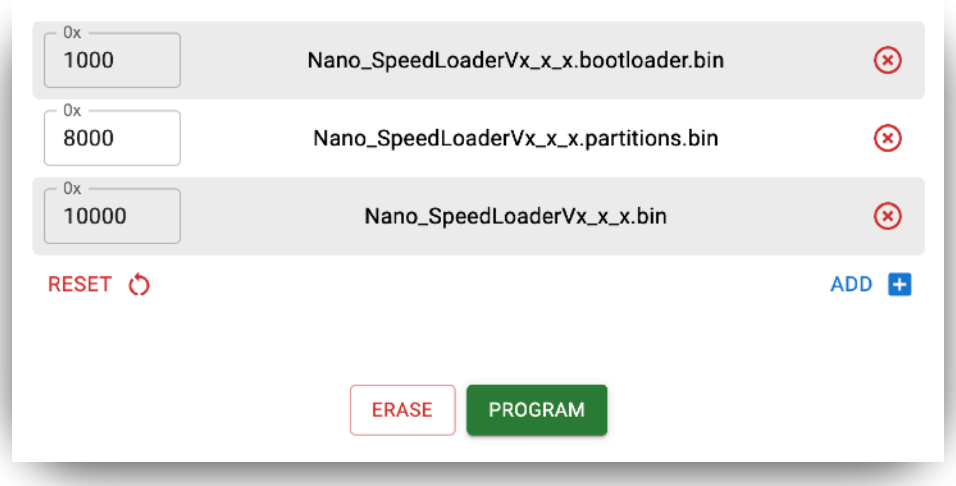
➡️ Connect your board to USB and click on **connect**. ⚠️ Make sure Arduino IDE is closed

➡️ Make sure each **address is properly entered** with corresponding file

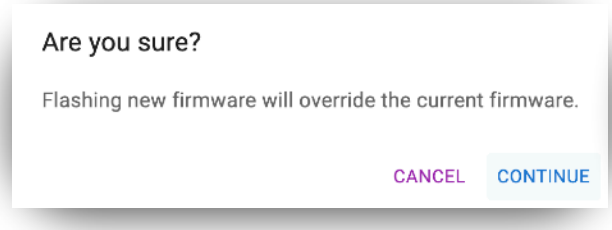
1000 : Nano_SpeedLoaderVx_x_x.bootloader.bin

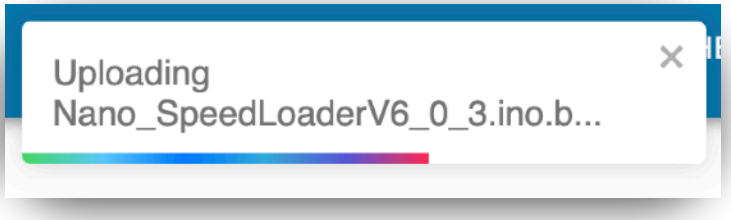
8000: Nano_SpeedLoaderVx_x_x.partitions.bin

10000: Nano_SpeedLoaderVx_x_x.bin (main firmware)

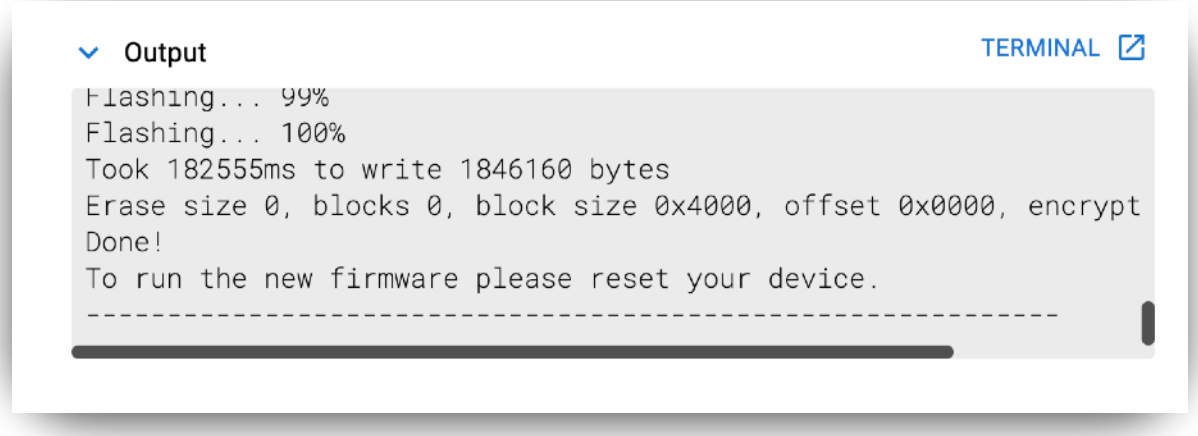


➡️ Press **PROGRAM**, and **confirm** overriding the firmware





ℹ️ Loading process takes up to **two minutes** for the very first upload. One the upload complete, simply unplug the board, and you are **ready** 🎉!



WIFI Updates (OTA) :

For future releases, Updating the board will be very easy and via WIFI, on the NANO,
➡️ go to : **Settings** > **About** > l**ong press on OTA**. (Follow instructions on the screen same as below)
ℹ️ The board will restart in **update mode**.

➡️ Using any device, connect to the **wifi network** named « NANO - UPDATE » and enter the password : « protoforge123 »

➡️ Go to URL : « **192.168.4.1** »

➡️ Select the firmware file and hit Upload, the board will take care for the rest then restart it self.

⚠️ Make sure to use the right firmware file (no boot loader neither partitions needed here) file to use : **`Nano_SpeedLoaderVx_x_x.bin (main firmware)`**

4 Assembly 2 hours

The following section describes how to assemble the 3D printed parts and electronic components all together, it is organized in two parts, that follow the recommended and easiest way to put everything together, for each part you have some additional informations and recommandations to help.

1) Gearbox-servo Assembly
2) Final Assembly

LIST OF HARDWARE REQUIRED FOR THIS BUILD:

	M3 Insert	M3*16	M3*6 (flat head)
Number	11	5	7

Gearbox - Servo Assembly:

Feeding wheel:



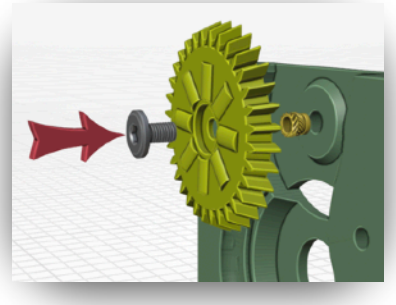
‼️ Make sure to remove the supports from the wheel and sand the surface to ensure flat surface for perfect fit

➡️ Install the Center screw M3*16mm to ensure servo horn alignment.

➡️ Install the round servo horn on the wheel using two of the screws provided in the servo package. The wheel has a notch to accommodate the horn.

ℹ️ Make sure the servo horn is straight with the wheel to ensure the servo axle is perpendicular to the wheel to avoid wobbling spin.

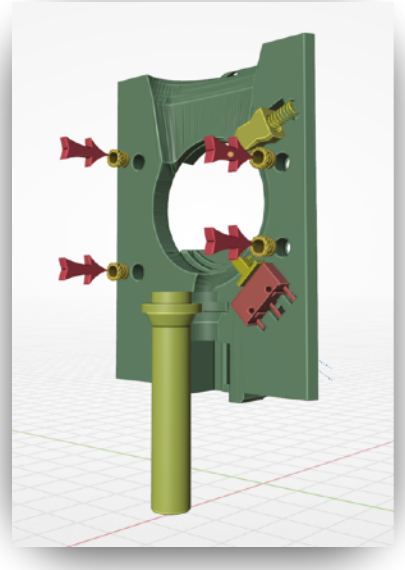
Stir wheel:



➡️ Install the **threaded insert** in its place then install the wheel using the **flat head** M3*6 screw and some **thread lock**.

ℹ️ Make sure the screw is **not over-tightened**, make few spins by hand to ensure the wheel turn freely **without wobbling**. Then let the thread lock dry few minutes.

‼️ As the wheel **turn clockwise** the crew will tighten and jam the whole assembly, make sure to **use some thread lock** to secure it in place.



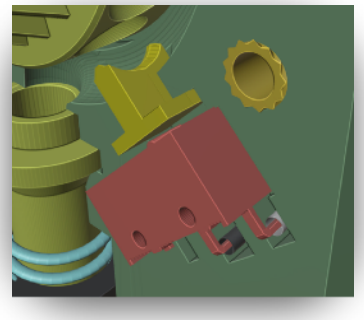
Gearbox Part 1:

➡️ Install the **threaded inserts** in the dedicated holes

➡️ Install the **Anti-pinch knob** using a pen spring.

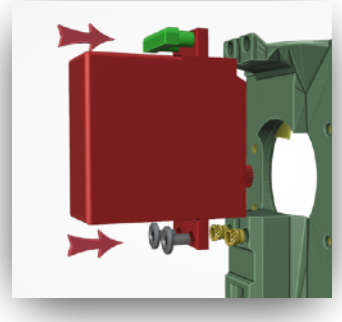
➡️ Install **the barrel** in it's dedicated notch

➡️ Slide **microswitch wires** through **the exit holes** before installing the JST-2 connector.



ℹ️ You might need **to bend** the counter micro switch pins in **L shape** to face the exit holes.

➡️ Counter **Knob** does not have orientation, it can be installed in **both orientations**.

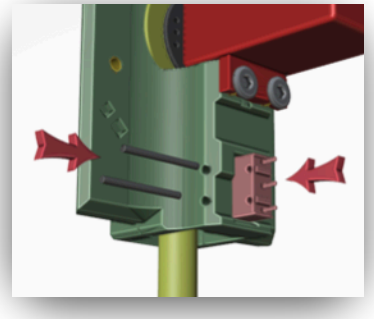


On the other side of the gearbox:

➡️ Install the two **2x M3 Heated inserts** in the l**ower holes**.

➡️ **Set the servo** in it's place using two flat head M3*6 screws (**do not tighten** before alignment).

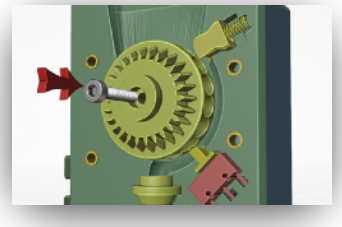
➡️ On the top side install the **servo clip**.

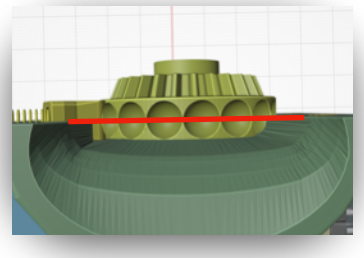


➡️ install the **trigger microswitch** then use some **3D printing filament** that slide through the hole to lock the microswitch in place.

ℹ️ test the **microswitch hole size** using pace 3D print filament, if too tight, you can enlarge it using a small 2mm drill bit carefully to not damage internal components.

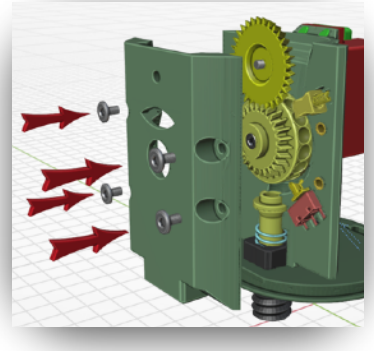
ℹ️ Before tightening the servo in place make sure that the wheel do not rub against the gearbox walls, spinning the wheel will help to find the best servo position. Various servo models can have few differences holes placement, use recommended servo to avoid issues.

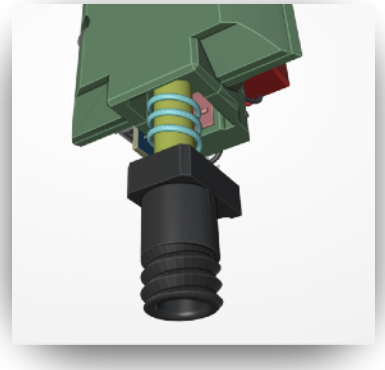




ℹ️ While tightening the wheel make sure to have the center of the wheel aligned with anti-pinch knob. If the wheel is not well centered, this might lead to unnecessary friction and power loss.

➡️ Close the Gearbox using four M3*6 Flat head screws that will maintain everything together.





➡️ Put the **trigger** and its **spring** in place , the **bottom plate** will come later to hold them in place.

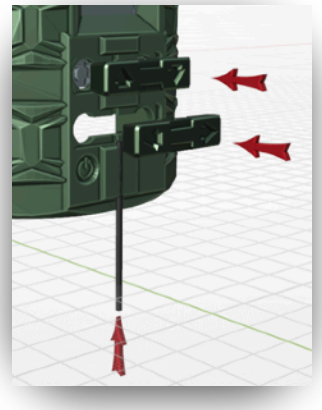
ℹ️ The spring can be **repurposed** from any where, however **avoid gearbox springs** as it's too strong, try finding some lower force. Perfect diameter from my tests is **0.7*10mm*10mm** spring.

Final Assembly :



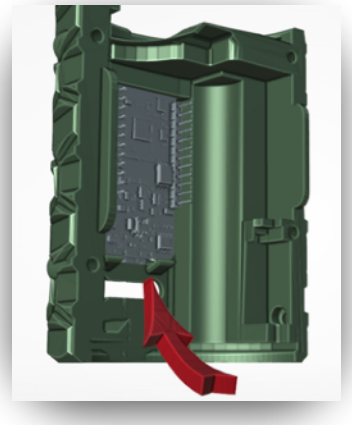
ℹ️ Make sure to have removed all support material from the front and back shells to let the assembly snap together easily, the assembly fit is adjusted, it allow some tolerance, so the inner surface wont require lot of preparation. You will need a Hobby knife, tweezers and pliers

Front Shell :



➡️ Install the buttons in place and secure them sliding a 3D print filament through the hole, then cut it to length.

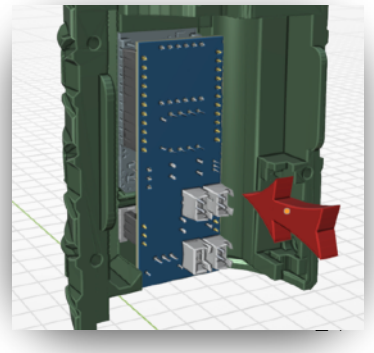
ℹ️ You might need to clean the hole using a 2mm drill bit in case the filament cannot make it's way through the channel because of the printing supports.



➡️ Install the Polycarbonate shield in the dedicated notch, use some drops of glue to secure it in place

➡️ On the inner side of the shell, there is a dedicated notch designed to receive the top end of the TTGO board. This notch helps guide and hold the board in place.

➡️ The board will get seated on it's dedicated place and secured with a small screw or a drop of superGlue



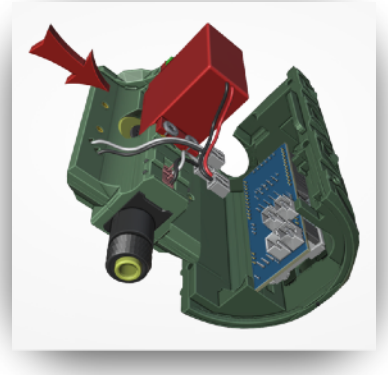
➡️ Plug in the Custom circuit board, it can be easily installed thanks to the header pins that allow easy install action and removal to the TTGO for programming.



‼ **Advanced Pack :** Please mind that Only 11 lower pins are used, make sure the two first pins are **not Plugged**.

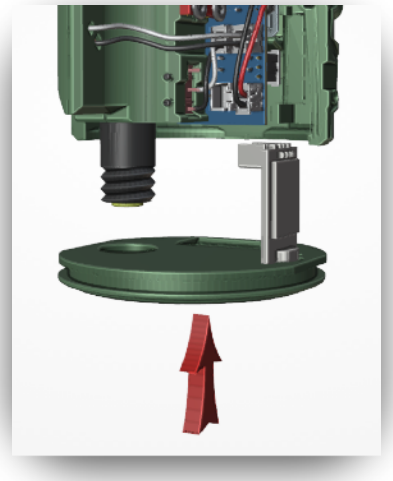
‼ **WARNING :** Make sure to have set the Regulator to the right voltage (5v) before to install the TTGO board

➡ Once the Board is in place, it is time to **install the servo-gearbox assembly** inside the front housing using the **dedicated rails** and notches, assembly should slide into place without effort.



➡ Using the **tweezers**, **plug the connectors** in place (servo, counter and trigger)

ℹ Advanced packs: each **connection is labelled** to avoid confusion.



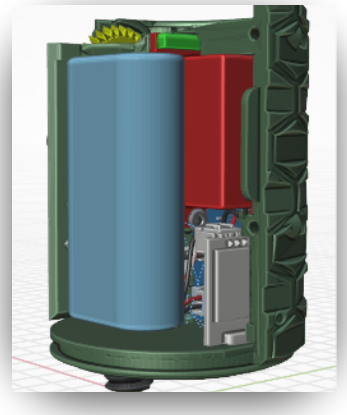
➡ Install the **BMS module**, the lower cover and the battery.

➡ **Plug battery connectors** to the dedicated connectors.

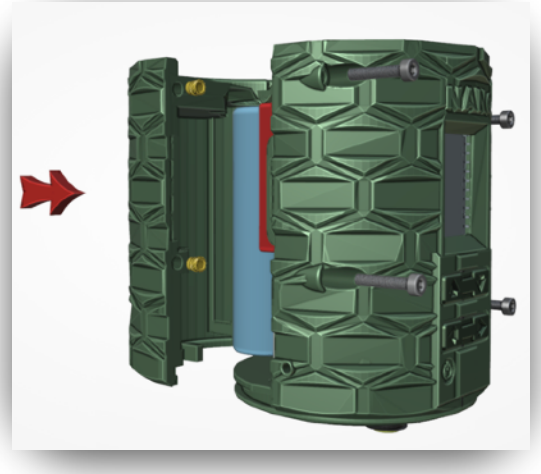
➡ **Install BMS** on the **bottom plate** then slide them inside the shell.

‼ BMS Module : Protect the back side of the JST-3 connector (pins) with some isolation tape, any shortcut will blow the IC instantly. (I Learn that the hard way).

ℹ Use some **tape** to maintain the **excess wire** around the battery, the objective is to keep the tight space **tidy** and **avoid creating pressure** on the board that might press the buttons.



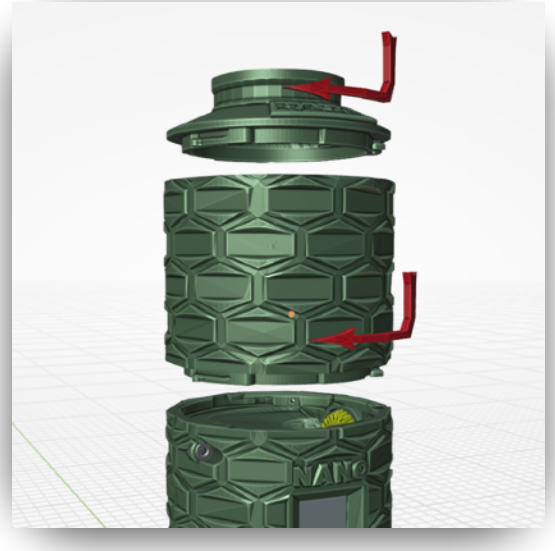
ℹ With some **sand paper**, you can sand the interior of the **trigger hole** and **bottom plate trigger hole**, this will reduce friction from the 3D printer layers



➡ Install four (4x) M3 heated insert on the back shell.

➡ Add the back shell to the assembly, it should slide in place easily. Add the four (4x) M3*16mm screws. If you feels resistance make sure there is no 3d printing support residue.

‼️ **Do not over-tighten the screws**, as it might snap the shell in two.

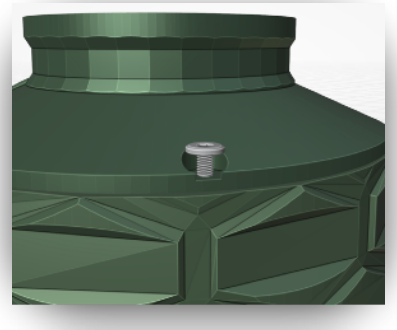


➡️ Last step is to **install the Extension** on the Speed loader, it can be fixed quickly thanks to the **L shape lock** mechanism.

➡️ The **top cover** can be added either directly on the Speedloader to have a minimalistic configuration or on top of the extension for full capacity.

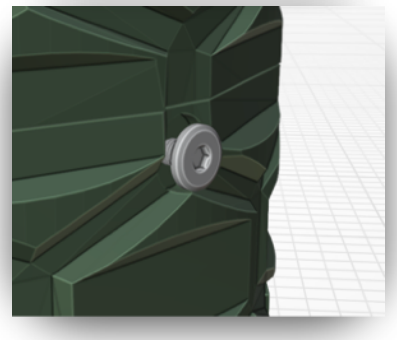


ℹ️ Use the **lock pictogram** to confirm that the mechanism is secured in place. Lock orientation is the same (clockwise) for the cover and the extension.

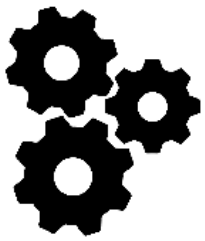


ℹ️ Both the extension and the top cover has a hole that allow to add a screw avoiding the loader to open.

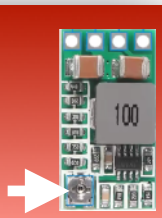
ℹ️ You can use the two screws left from the Servo package to secure the assembly, there is no heat insert required here, just the metal tapper screw is enough.



5 Break-in 2+ hours



WARNING /!\ Using a Multimeter, make sure to set the DC-DC regulator potentiometer to 5V before Plugging the TTGO otherwise you might burn the board.



Break-In Procedure

The break-in phase is designed to ensure that all components of your Nano Speedloader are properly aligned and running smoothly before regular use. During this phase, you will gradually test the mechanism in different conditions: first without BBs, then with BBs but no magazine, and finally with a fully connected magazine. This process allows the moving parts to settle into their optimal positions, while also giving you the chance to identify and correct any small issues early.

Dry Run (Without Magazine)

- Fill the reservoir with a small number of BBs.
- Run the speedloader to check how the feeding wheels handle the BBs.
- Observe whether BBs move consistently through the feeding channel without jamming.

ℹ️ It is normal for the servo to become louder during this test. The feeding barrel is designed with a **tight fit** to prevent BBs from simply falling out by gravity when no magazine is attached.

Crackling or Loud Clicking Noises : These sounds indicate that BBs are not feeding smoothly and may be encountering excessive friction or misalignment inside the channel.

Resolutions:

➡️ Check wheel alignment: Ensure the feeding wheel is perfectly perpendicular to the motor shaft. If it wobbles, BBs may rub against the gearbox walls.

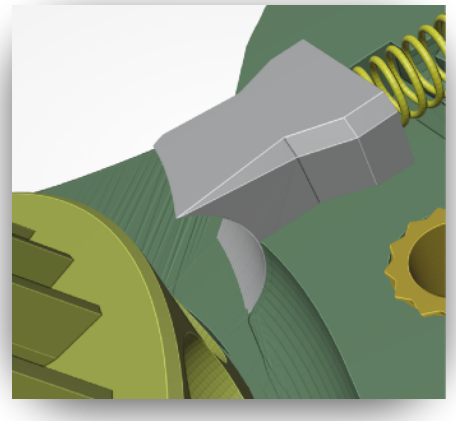
➡️ Visual alignment check: In the assembly guide, refer to the picture showing how the wheel should align with the anti-pinch knob when tightened.

➡️ Center screw adjustment: Some servos are sensitive to the pressure of the central screw. Try loosening it slightly, or even removing it temporarily, to confirm that the wheel spins freely without restrictions.

Knob Misalignment (Anti-Pinch or Counter Knob)

- A loud clicking can also result from one or both knobs being slightly misaligned.
- Although designed to be self-aligned, tolerances from your printer, filament, or print settings may introduce a small offset.

Resolutions:



➡️ **Adjust with sandpaper:** Lightly sand the affected knob surfaces to correct their alignment.

➡️ **Confirm counter knob function:** The counter knob should be pushed by each passing BB, but the passage should not create noticeable resistance or friction.

➡️ **Refer to close-up picture:** In the manual, check the provided illustration to confirm the proper alignment.

Inconsistent Counting

- Test the counter by emptying the speedloader and loading an exact number of BBs (minimum of 10 to fully engage the feeding wheel).
- Run the speedloader on an unlimited preset and verify the screen displays the correct number.
- Reset the counter and repeat several times to confirm reliability.

🔁 If the displayed count varies between tests, the feeding wheel and servo are not perfectly centered. Misalignment causes the wheel to move closer/farther from the gearbox wall during rotation, preventing some BBs from pressing the counter knob.

Resolutions:

➡️ **Go back to Step 1:** Revisit the "**Initial Dry Run**" and realign the servo. Loosen the screws, adjust the position, and retest until the counting is accurate and repeatable.

Step 2 – BB Run (With Magazine)

This step ensures the speedloader is fully capable of handling real-world conditions and heavier loads. After confirming that the mechanism works smoothly without a magazine, it's time to test the complete feeding cycle with an actual magazine.

1. **Preset Selection**
    - Choose a preset that matches the capacity of your magazine.
    - A mid-cap magazine is ideal for this test, with a typical capacity of **120–160 BBs**. This range is perfect for running the loader while continuing the break-in process.
2. **Running the Test**
    - Run approximately **2000 BBs** through the same magazine by repeatedly loading and unloading it.
    - Compare the number of BBs fed during each refill. Consistency across refills confirms the **counting system accuracy**.
3. **Expected Behavior**
    - As the magazine fills, it is **normal to hear the servo struggling more and more**.
    - This is due to the increasing back-pressure from the spring inside the magazine, which resists further BBs being loaded.

Common issues :

2- if the motor is turning but the bb are not loading, it might be an inversion of the two motor wiring :

➡️ go to settings > Servo > then adjust the servo direction CW<>CCW

3- if nothing happens when pressing the Trigger, it is most probably because the connectors are inverted.

➡️Check if all connectors are in their dedicated places.

4. Nothing happen when pressing the power button: it might come from several parts,

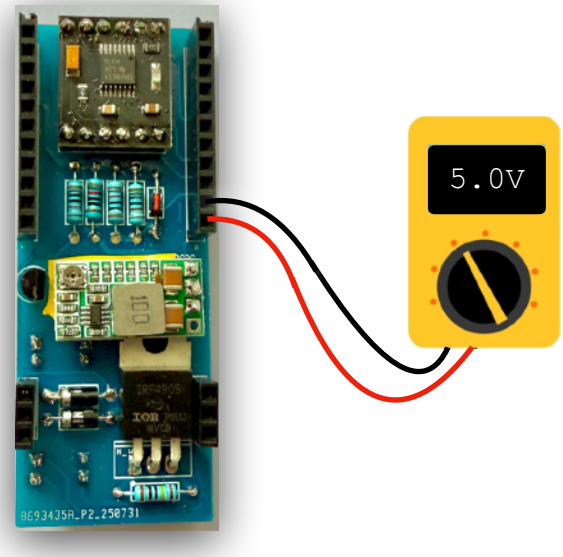
Resolution:

➡️ Make sure you have Wired the battery in the right polarity (Page 19)

➡️ Check the soldering of the components and the modules

➡️ Make sure you have set the voltage to 5v using the potentiometer otherwise the board will restart in loop of even be damaged.

↪️ the multimeter is your best FRIEND, along the way of soldering the wires to make sure everything is linked



6. Project Limitations



While this upgrade pack brings several enhancements, there are still limitations and areas for improvement. Community feedback is highly valued to refine the design and functionality.

Current Limitations:

- **3D-Printed Parts Durability:** Some components may experience wear over time, especially under high-speed operation. Thus it is recommended to print parts using high strength filaments like PETG or ABS (I choose this option)
- **Speed Constraints:** Loading speeds beyond 25 BBs per second may risk breaking the 3D-printed components.

Ongoing Issues & Areas for Improvement:

- **Fine-Tuning Motor Speed:** Further refinements in the motor controller logic could improve smoothness and efficiency. However, reducing the motor speed also decreases its torque. If the speed is set too low, the servo motor may lack the strength to load magazines with stronger springs. To address this issue, increasing the motor speed may be necessary to ensure proper functionality.
- **User Interface Enhancements:** The onboard software is developed on Arduino to provide advanced features and an improved user experience. The software development started in May 2024 and has taken more than two months to reach a proper running state. After the first version of the software, multiple features have been added, such as the ability to rename each preset, backlight control, board authority over the entire circuit, and a loading timer. The software continues to evolve based on real-world Airsoft field usage and community feedback. More customization options for display elements could further enhance usability.
- **Firmware Stability:** Continuous software updates are required to address bugs and optimize performance.

How You Can Help:

Community contributions in the form of feedback, suggested improvements, and testing are welcome. If you have ideas or solutions for any of the above issues, feel free to share them on our Discord/forum or through direct communication.

7. Additional Notes & Support

Maintenance & Longevity:

- Periodically check for wear on moving parts.
- Clean and lubricate where necessary to ensure smooth operation.
- Avoid excessive force or stress on 3D-printed components.
- Remove the battery if the Box is stored unused for more than two weeks

Community & Support:

- Join our Discord/forum for troubleshooting and user discussions.
- Check for updates and new versions on Cults3D.
- Feedback is welcome! Share your modifications and experiences with the community.
- You can reach out to me via email on yaceairsoftcreations@gmail.com

This guide ensures you can successfully install and utilize this upgrade pack for enhanced performance. Happy assembling!